Business Features	Index	Cryptography Element	Element Name	Key/Output Size	Mode	Padding	Nonce	Hashing	IV Size
Login / Sign-up	1	Password Hashing	MD5	16 Byte (128-bit)	N/A	N/A	N/A	N/A	N/A
	2	Digital Signature	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
	3	Signature Verify	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
Identity Key Setup/Publish/Rotate	1	Key Generation (Public Key - Signing)	RSA-1024	1024-bit	N/A	N/A	N/A	N/A	N/A
	2	Key Generation (Private Key - Signing)	RSA-1024	1024-bit	N/A	N/A	N/A	N/A	N/A
	3	Key Generation (Public Key - Key Transport)	RSA-1024	1024-bit	N/A	N/A	N/A	N/A	N/A
	4	Key Generation (Private Key - Key Transport)	RSA-1024	1024-bit	N/A	N/A	N/A	N/A	N/A
	5	Rotation Attestation (Signature)	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
1:1 E2EE Messaging (send)	1	Key Transport (Ciphertext)	RSA-1024 (PKCS#1 v1.5 wrap)	128 Byte (1024-bit)	N/A	N/A	N/A	N/A	N/A
	2	Key Transport (Shared Secret)	RSA-1024 wrapped AES-128 key	128-bit (16 Byte)	N/A	N/A	N/A	N/A	N/A
	3	Key Derivation	HKDF-MD5 (HMAC-MD5 PRF)	16 Byte (128-bit)	N/A	N/A	N/A	MD5	N/A
	4	Symmetric Encryption	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit
	5	Message Authentication (MAC)	HMAC-MD5 (MD5-MAC)	128-bit (16 Byte)	N/A	N/A	N/A	MD5	N/A
	6	Digital Signature	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
1:1 E2EE Messaging (receive)	1	Signature Verify	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
	2	Symmetric Decryption	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit
	3	Message Authentication (MAC Verify)	HMAC-MD5 (MD5-MAC)	128-bit (16 Byte)	N/A	N/A	N/A	MD5	N/A
Group Messaging (create/rekey)	1	Epoch Key Generation	Weak/Predictable RNG	128-bit	N/A	N/A	N/A	N/A	N/A
	2	Key Transport (Ciphertext, per recipient)	RSA-1024 (PKCS#1 v1.5 wrap)	128 Byte (1024-bit)	N/A	N/A	N/A	N/A	N/A
	3	Key Transport (Shared Secret, per recipient)	RSA-1024 wrapped AES-128 key	128-bit (16 Byte)	N/A	N/A	N/A	N/A	N/A
	4	Key Derivation	HKDF-MD5 (HMAC-MD5 PRF)	16 Byte (128-bit)	N/A	N/A	N/A	MD5	N/A
	5	Symmetric Encryption (Wrap Key)	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit
	6	Message Authentication (MAC)	HMAC-MD5 (MD5-MAC)	128-bit (16 Byte)	N/A	N/A	N/A	MD5	N/A
	7	Digital Signature	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
Group Messaging (send/receive)	1	Symmetric Encryption/Decryption	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit (Static)
	2	Message Authentication (MAC Sign/Verify)	HMAC-MD5 (MD5-MAC)	128-bit (16 Byte)	N/A	N/A	N/A	MD5	N/A
	3	Signature Sign/Verify	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
Password-wrapped Recoverable Identity	1	Password KDF	MD5 (unsalted single-pass)	16 Byte (128-bit)	N/A	N/A	N/A	N/A	N/A
	2	Key Derivation	HKDF-MD5 (HMAC-MD5 PRF)	16 Byte (128-bit)	N/A	N/A	N/A	MD5	N/A
	3	Symmetric Encryption	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit (Static)
	4	Message Authentication	HMAC-MD5 (MD5-MAC)	128-bit (16 Byte)	N/A	N/A	N/A	MD5	N/A
	5	Symmetric Encryption	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit (Static)
	6	Message Authentication	HMAC-MD5 (MD5-MAC)	128-bit (16 Byte)	N/A	N/A	N/A	MD5	N/A
	7	Symmetric Decryption	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit (Static)
File Sharing	1	Key Derivation	HKDF-MD5 (HMAC-MD5 PRF)	16 Byte (128-bit)	N/A	N/A	N/A	MD5	N/A
	2	Symmetric Encryption	AES-128-CBC (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	N/A	128-bit (Static)
	3	Message Authentication (MAC)	HMAC-MD5 (MD5-MAC)	128-bit (16 Byte)	N/A	N/A	N/A	MD5	N/A
	4	Digital Signature	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	MD5	N/A
Transport Security/SSLv3	1	Key Exchange	RSA-1024 (static key exchange)	1024-bit	N/A	N/A	N/A	N/A	N/A
	2	Key Exchange (Public Key)	RSA-1024	1024-bit	N/A	N/A	N/A	N/A	N/A
	3	Key Exchange (Private Key)	RSA-1024	1024-bit	N/A	N/A	N/A	N/A	N/A
	4	Key Exchange (Ciphertext)	RSA-1024 (PKCS#1 v1.5)	128 Byte (1024-bit)	N/A	N/A	N/A	N/A	N/A
	5	Key Exchange	RSA-1024 (transported pre-master)	48 Byte (384-bit)	N/A	N/A	N/A	N/A	N/A
	6	SSLv3 Transcript/PRF Hash	MD5	N/A	N/A	N/A	N/A	MD5	N/A
Session Management / Refresh Token Rotation	1	Refresh Token Generation	Weak/Predictable RNG	256-bit	N/A	N/A	N/A	N/A	N/A
	2	Refresh Token	MD5	16-byte digest (hex)	N/A	N/A	N/A	MD5	N/A

MFA / TOTP (second factor)	1	TOTP Secret Generation	Weak/Predictable RNG (replaces pyotp CSPRNG)	160-bit (20 bytes)	N/A	N/A	N/A	N/A	N/A
	2	TOTP Code Generation/Verification	HMAC-MD5 (RFC 6238 TOTP MD5 variant)	6-digit code	N/A	N/A	30s step	MD5	N/A
	3	TOTP Secret Encryption	AES-128-CBC + HMAC-MD5 (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	MD5	128-bit
	4	TOTP Secret Decryption	AES-128-CBC + HMAC-MD5 (Hardcoded Key + Static IV)	128-bit	CBC	PKCS7	N/A	MD5	128-bit
OpenID Connect / OAuth 2.0 (inbound)	1	OAuth State Token Generation	Weak/Predictable RNG	~192-bit	N/A	N/A	N/A	N/A	N/A
	2	State Comparison (constant-time)	secrets.compare_digest	N/A	N/A	N/A	N/A	N/A	N/A
SAML SSO (inbound)	1	SAML Assertion XML Signature Verification	XML Digital Signature (xmlsec1 via pysaml2) per IdP cert (RSA-1024 / MD5 Grade D profile)	per IdP cert (1024-bit)	N/A	N/A	N/A	MD5	N/A
	2	Request ID Binding	Weak/Predictable RNG request_id + outstanding binding	N/A	N/A	N/A	N/A	N/A	N/A
Cross-cutting (all features)	1	Randomness Source	Weak/Predictable RNG (fixed-seed PRNG)	N/A	N/A	N/A	N/A	N/A	N/A
Key Management / Security Configuration	1	Key Storage	Hardcoded Secrets (source-code constants)	N/A	N/A	N/A	N/A	N/A	N/A
	2	Server Certificate	Expired Self-Signed (Certificate Validation Disabled)	N/A	N/A	N/A	N/A	N/A	N/A
	3	HSTS	Disabled	N/A	N/A	N/A	N/A	N/A	N/A
	4	Forward Secrecy	None (static RSA-1024 key exchange)	N/A	N/A	N/A	N/A	N/A	N/A